

customer churn for a saas company via logistic regression

```
ubuntu@ubuntu:~/Documents/datascience/ML/churn_prediction_project_saas$ python3 main_logisticregression.py
  CustomerID  Age  Gender  SubscriptionType  MonthlyUsageHours  SupportTickets  Churn
0  CUST0000   56  Female          Basic             28.9              2        1
1  CUST0001   69  Female          Basic             36.4              1        0
2  CUST0002   46   Male    Enterprise             61.0              2        0
3  CUST0003   32  Female          Pro             43.3              1        1
4  CUST0004   60  Female          Pro             37.5              2        0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            1000 non-null   object
1   Age                   1000 non-null   int64
2   Gender                1000 non-null   object
3   SubscriptionType       1000 non-null   object
4   MonthlyUsageHours      1000 non-null   float64
5   SupportTickets         1000 non-null   int64
6   Churn                  1000 non-null   int64
dtypes: float64(1), int64(3), object(3)
memory usage: 54.8+ KB
None
Missing values:
  CustomerID      0
  Age             0
  Gender          0
  SubscriptionType 0
  MonthlyUsageHours 0
  SupportTickets  0
  Churn           0
dtype: int64
Confusion Matrix:
[[229  0]
 [ 71  0]]
```

```
Classification Report:
              precision    recall  f1-score   support

     0       0.76      1.00      0.87        229
     1       0.00      0.00      0.00         71

   accuracy      0.38
  macro avg      0.38      0.50      0.43        300
weighted avg      0.58      0.76      0.66        300
```

ROC AUC Score: 0.6795620886893412

Top Contributing Features:

	Feature	Coefficient
4	SubscriptionType_Enterprise	-2.027570
5	SubscriptionType_Pro	-1.668972
3	SupportTickets	0.071376
2	MonthlyUsageHours	-0.034997
0	Age	0.024777
1	Gender	-0.019072

/home/ubuntu/.local/lib/python3.10/site-packages/matplotlib/projections/_init_.py:63: UserWarning: Unable to import Axes3D. This may be due to multiple versions of Matplotlib being installed (e.g. as a system package and as a pip package). As a result, the 3D projection is not available.

warnings.warn("Unable to import Axes3D. This may be due to multiple versions of "

Prediction results saved to 'churn_logistic_predictions.csv'

```
data_generator.py
main_logisticregression.ipynb
main_logisticregression.py
main_RandomForestClassifier.ipynb
main_RandomForestClassifier.py
saas_customer_data.csv

# ===== 1. Load and Inspect Data =====
import pandas as pd

df = pd.read_csv('saas_customer_data.csv')
print(df.head())
print(df.info())
print("Missing values:\n", df.isnull().sum())

# ===== 2. Data Preprocessing =====
from sklearn.preprocessing import StandardScaler

# Drop CustomerID (non-informative)
df.drop(columns=['CustomerID'], inplace=True)

# Encode Gender
df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})

# One-hot encode SubscriptionType
df = pd.get_dummies(df, columns=['SubscriptionType'], drop_first=True)

# Separate features and target
X = df.drop('Churn', axis=1)
y = df['Churn']

# Scale numeric features
scaler = StandardScaler()
X[['Age', 'MonthlyUsageHours', 'SupportTickets']] = scaler.fit_transform(
    X[['Age', 'MonthlyUsageHours', 'SupportTickets']]
)
```

OUTLINE

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# ===== 3. Train/Test Split =====
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=42
)

# ===== 4. Model Training (Logistic Regression) =====
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score, roc_curve

logreg = LogisticRegression(max_iter=1000, solver='liblinear')
logreg.fit(X_train, y_train)

y_pred = logreg.predict(X_test)
y_proba = logreg.predict_proba(X_test)[:, 1]

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("ROC AUC Score:", roc_auc_score(y_test, y_proba))
```

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# ===== 5. Interpret Model Coefficients =====
coeff_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': logreg.coef_[0]
}).sort_values(by='Coefficient', key=abs, ascending=False)

print("\nTop Contributing Features:\n", coeff_df)

# Visualize coefficients
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(x='Coefficient', y='Feature', data=coeff_df)
plt.title("Logistic Regression Coefficients")
plt.show()

# ===== 6. ROC Curve Visualization =====
fpr, tpr, thresholds = roc_curve(y_test, y_proba)

plt.figure(figsize=(8, 5))
plt.plot(fpr, tpr, label=f'AUC = {roc_auc_score(y_test, y_proba):.2f}')
plt.plot([0, 1], [0, 1], '--', color='gray')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend()
plt.show()

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# ===== 7. Export Results for Stakeholders =====
results_df = X_test.copy()
results_df['ActualChurn'] = y_test
results_df['PredictedChurn'] = y_pred
results_df['ChurnProbability'] = y_proba

results_df.to_csv('churn_logistic_predictions.csv', index=False)
print("Prediction results saved to 'churn_logistic_predictions.csv'")
```

	CustomerID	Age	Gender	SubscriptionType	MonthlyUsageHours	SupportTickets	\
0	CUST0000	56	Female	Basic	28.9	2	
1	CUST0001	69	Female	Basic	36.4	1	
2	CUST0002	46	Male	Enterprise	61.0	2	
3	CUST0003	32	Female	Pro	43.3	1	
4	CUST0004	60	Female	Pro	37.5	2	

Churn

0	1
1	0
2	0
3	1
4	0

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	CustomerID	1000 non-null	object
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2	Gender	1000 non-null	object

Churn

0 1

1 0

2 0

3 1

4 0

<class 'pandas.core.frame.DataFrame'>

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Data columns (total 7 columns):

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4	MonthlyUsageHours	1000 non-null	float64
5	SupportTickets	1000 non-null	int64
6	Churn	1000 non-null	int64

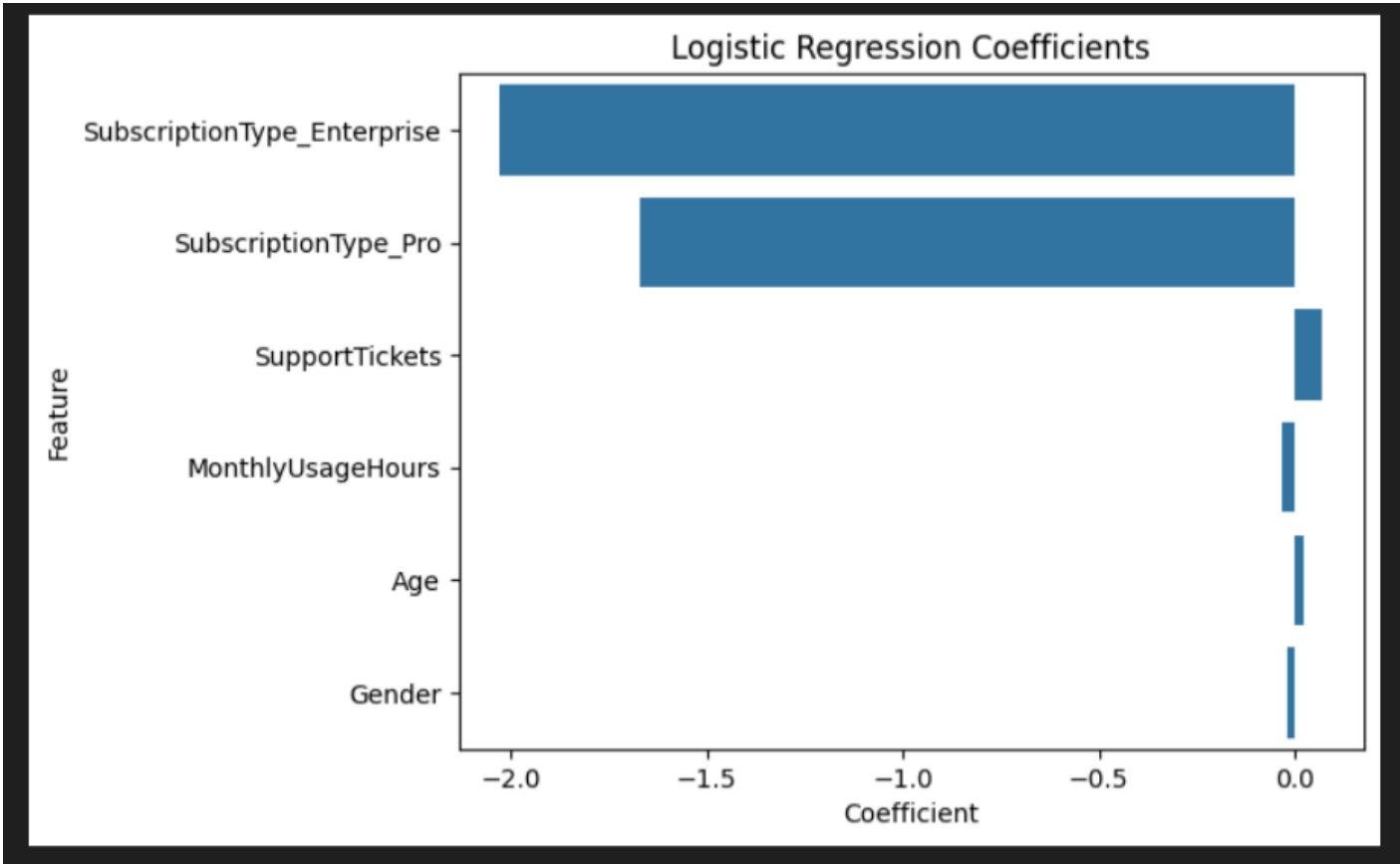
...

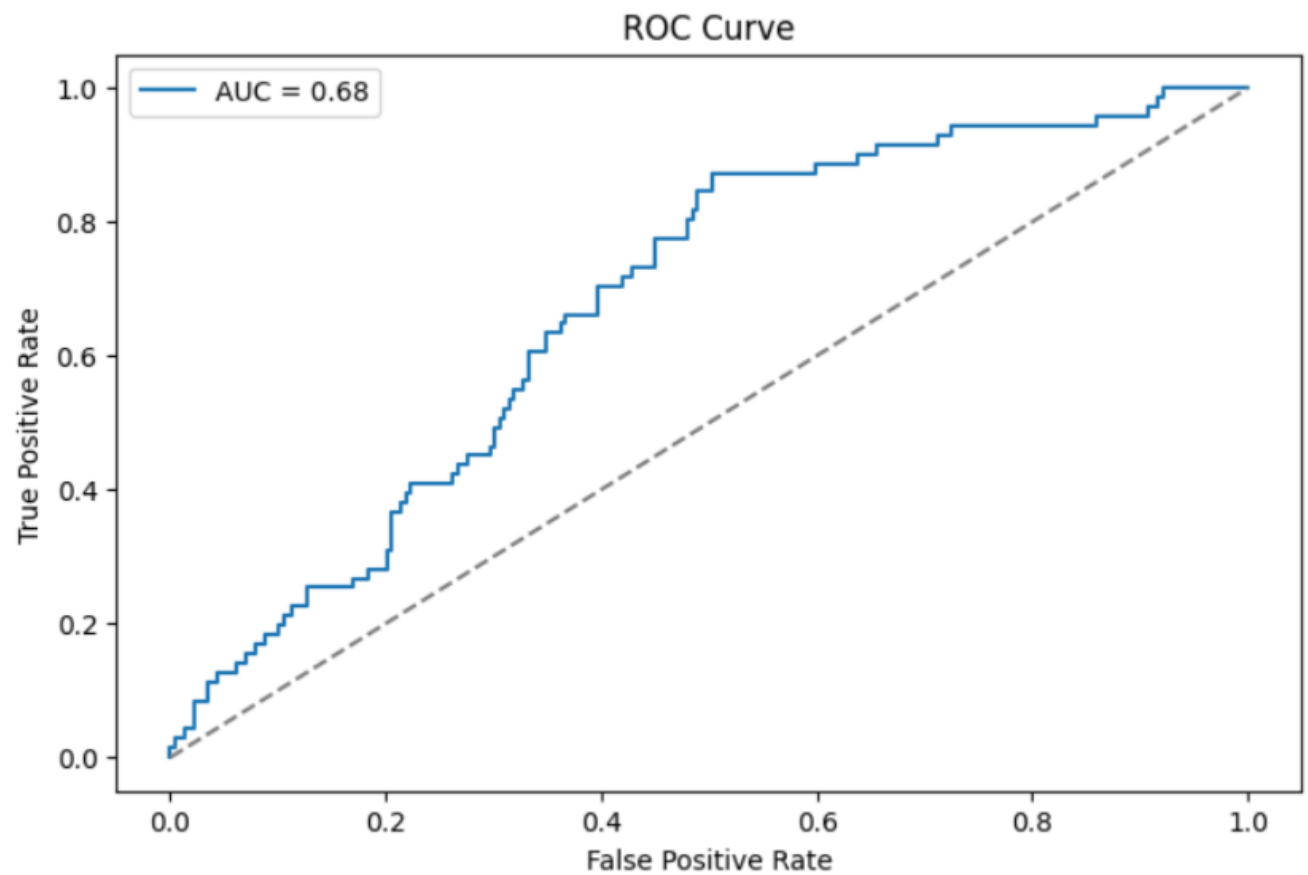
3 SupportTickets 0.071376

2 MonthlyUsageHours -0.034997

0 Age 0.024777

1 Gender -0.019072





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